Outlook

Decision: MAJOR REVISION (RQ) - T-IFS-26361-2026

From Transactions on Information Forensics & Security <onbehalfof@manuscriptcentral.com>
Date Wed 14 Oct 2026 03:44
To Liu Renyang <ryliu@nus.edu.sg>
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- External Email -

13-Oct-2026

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Paper:T-IFS-26361-2026, "CAAP: Capture-Aware Adversarial Patch Attacks on Palmprint Recognition Models"

Dear Dr. Renyang Liu,

I am writing to you concerning the above referenced manuscript, which you submitted to the IEEE Transactions on Information Forensics & Security.

Based on the enclosed set of reviews, your manuscript requires a MAJOR REVISION (RQ).

Associate Editor's Comments:
The manuscript addresses a relevant biometric security problem and presents promising digital results. However, substantial clarification and additional experimental validation are needed to support the claims of physically realizable attacks. Please address all reviewer comments, with particular attention to the consistency between input-conditioned rendering and physical deployment, the adequacy of the physical evaluation, and the clarity of the biometric evaluation protocol.

When you are ready to submit your revision, visit the following link: https://ddec1-0-en-ctp.trendmicro.com:443/wis/clicktime/v1/query?url=https%3a%2f%2fieee.atyponrex.com%2fsubmission%2fsubmissionBoard%2fREX%2dPROD%2d2%2d8D394CC2%2dDAE5%2d4E0A%2d8BCC%2d8A053465A475%2dD18E32F9%2d4A85%2d420C%2dA165%2d8901C785F73A%2d80859%2fcurrent%3fidtype%3dexternal&umid=57e14d48-c20b-40ed-80fc-a874e88664fd&rct=1791920635&auth=8d3ccd473d52f326e51c0f75cb32c9541898e5d5-be9108302fcecd671b26eb00eacab29bce49754e

Your revised manuscript must be submitted back no later than 6 weeks from the date of this letter together with a required point-by-point reply that explains how you addressed the reviewers' comments. If we do not receive your revised manuscript within 6 weeks from the date of this letter, your manuscript will be considered withdrawn.

When submitting your revised manuscript, please make sure to also upload your required Author Response file.

If you have any questions regarding the reviews, please contact me. Any other inquiries should be directed to Rida Sohail.

NOTE – Open Access:
The publication is a hybrid journal, allowing either Traditional manuscript submission or Open Access (author-pays OA) manuscript submission. Upon submission of your final files, if you choose to have your manuscript be an Open Access article, you commit to pay the discounted OA fee if your manuscript is accepted for publication in order to enable unrestricted public access. As of 01 January 2017, the OA fee is $1,950. Any other application charges (such as charge for the use of color in the print format) will be billed separately once the manuscript formatting is complete but prior to the publication. If you would like your manuscript to be a Traditional submission, your article will be available to qualified subscribers and purchasers via IEEE Xplore. No OA payment is required for Traditional submission.

NOTE – Overlength manuscripts:
The final submitted version of your manuscript may not exceed 16 pages in the double-column, single-spaced format including figures, photos, and bios. If your paper is accepted, it will not be sent to production if it exceeds this limit. Additionally, any manuscript that exceeds 10 pages in double-column format will be charged mandatory overlength fees. The rate is $220/page starting on page 11 of a published manuscript.

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Best regards,

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Reviewer Comments:

Reviewer: 1

Comments:
This paper proposes CAAP, a capture-aware adversarial patch framework for physically realizable attacks on deep palmprint recognition systems. The technical contributions are solid, and the experimental setup is comprehensive. Extensive experiments across three datasets and multiple victim architectures demonstrate strong attack performance, cross-model transferability, and physical-world feasibility. However, several issues need to be addressed to improve the quality of this manuscript.

1.The core novelty is limited. The cross-shaped patch topology is not original to this work, it was proposed by Ran et al. (TCSVT 2024), and the paper simply adopts it for the palmprint domain.
2.The main contributions (ASIT, RaS, MS-DIFE) are design choices rather than fundamental methodological breakthroughs.
3.In Eq.(4) of Section IV.C, the authors claimed that \phi is a learnable parameter. However, they does not indicate its specific value.
4.In Eq.(7), what do c and b stand for? Are they learnable? If so, what are the final parameters? The author should state how to select these two parameters.
5.Moreover, Sections III and IV introduce many variables and equations that are difficult to follow. The authors should provide a notation table to review all the symbols used in this article.
6.Some abbreviations should be given their full names when they first appear so that readers can follow them better.
7.This paper employs the CAAPc and CAAPs variants. The author should clarify how to choose between the two variants and what application scenarios each is suitable for.

Additional Questions:
Is the work within the scope of the journal?: Clearly within scope

Is the manuscript technically correct?: Some minor concerns that should be easily corrected without altering the contribution or conclusions

Is the technical contribution novel?: Moderate novelty, with clear extensions of existing methods/concepts

Is the technical contribution significant?: Substantial contribution, with a clear potential for impact

Is the length of the manuscript appropriate to the contribution?: Length is appropriate

Are the references appropriate, without any significant omissions?: Complete list of references without any significant omissions

Are there any references that do not appear to be relevant?: All references are directly relevant to the contribution of the manuscript

Is the manuscript properly structured and clearly written?: Some minor structural, language, or other issues of exposition that would be easily rectified

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

: no

Reviewer: 2

Comments:

This paper proposes CAAP, a universal adversarial patch attack for deep palmprint recognition. It combines a cross-shaped mask, an input-conditioned spatial transformer, stochastic radiometric capture simulation, and multi-scale feature guidance. Experiments report strong untargeted and targeted attack success, cross-model/cross-dataset transfer, partial resistance to adversarial training, and preliminary print-and-capture effectiveness.

1. ASIT requires a clean ROI to predict sample-specific geometric and photometric transformations before digitally compositing the patch. It is unclear how an attacker using a fixed printed patch could obtain this input or realize these adaptive transformations. The physical deployment procedure must be clarified and evaluated without digital test-time manipulation.

2. The physical evaluation is insufficient. Only ten subjects and a limited acquisition setup are used. Moreover, an identity is considered successfully attacked if just one of 20 attempts succeeds, which can substantially inflate the reported ASR. The paper should report per-attempt ASR, per-subject distributions, confidence intervals, and results across different devices, poses, distances, lighting conditions, and patch placements.

3. The biometric evaluation protocol is unclear. The claimed subject-disjoint split appears inconsistent with closed-set identity classification and a fixed target label. The authors should clearly describe victim-model training, patch-training, enrollment, gallery, and probe partitions. Since the paper motivates authentication applications, verification metrics such as false-accept rate and ROC/DET curves should also be reported.

4. Essential settings, including ASIT bounds, RaS distributions, EOT samples, training iterations, MS-DIFE pretraining, and exact patch areas—are missing. It is also unclear whether all baselines receive equal optimization budgets, transformation models, and tuning.

5. The novelty and component contributions require stronger evidence. Cross-shaped patches, EOT, spatial transformers, and feature-space losses are established techniques. The ablation results suggest that ASIT provides most of the improvement, while RaS and MS-DIFE contribute little individually. In particular, the physical benefit of RaS should be demonstrated through controlled physical ablations.

Additional Questions:
Is the work within the scope of the journal?: Clearly within scope

Is the manuscript technically correct?: Moderate concerns with the potential for some impact on the contribution or conclusions

Is the technical contribution novel?: Moderate novelty, with clear extensions of existing methods/concepts

Is the technical contribution significant?: Moderate contribution, with the possibility of an impact on the field

Is the length of the manuscript appropriate to the contribution?: Length is appropriate

Are the references appropriate, without any significant omissions?: Complete list of references without any significant omissions

Are there any references that do not appear to be relevant?: All references are directly relevant to the contribution of the manuscript

Is the manuscript properly structured and clearly written?: Some minor structural, language, or other issues of exposition that would be easily rectified

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.
: NA

Reviewer: 3

Comments:
The paper proposes CAAP, a universal adversarial patch framework for palmprint recognition combining a cross-shaped mask, input-conditioned rendering, stochastic capture augmentation, and multi-scale identity-feature guidance. I find the problem relevant to biometric security and appreciate the evaluation across three datasets and seven recognizers. The digital untargeted results are promising, and the transfer experiments and physical captures provide useful initial evidence. However, several aspects require substantial clarification and validation.

1. My main concern is the correspondence between digital rendering and physical deployment. Sections IV-C and IV-G, including Eqs. (4), (8), and (23)–(24), use each clean ROI to predict placement, scale, and patch-local contrast and brightness, whereas Section III-C describes fabricating and reusing one physical patch. Please explain how the attacker obtains the clean ROI and physically implements these adjustments with the same printed artifact. If ASIT instead simulates acquisition, its optimized transformations should be distinguished from externally imposed capture variation.
2. The physical success metric needs more complete reporting. Section V-F and Fig. 5 count a subject as successfully attacked if any of twenty captures succeeds. Thus, the 90% untargeted result on CCNet means nine of ten subjects succeeded at least once. This is meaningful for a stated retry budget, but per-capture success rates and each subject's success counts are also needed. Please include clean recapture and same-shape occlusion controls, a matched universal cross-shaped EOT patch, and physical ASIT/RaS ablations under specified lighting and pose variation. These comparisons are necessary to assess the physical benefit of the capture-aware design.
3. Section V-A states that training and test subjects are disjoint, yet the attack uses identity-class logits and reports near-saturated clean classification accuracy. Please distinguish recognizer training, enrollment, patch training, and evaluation, and explain how test identities enter the classifier or gallery. Clean accuracy and the ASR denominator should also be reported.
4. I find the coverage of recent palmprint research insufficient, particularly relevant work published over the past two years in TIFS, TIP, and major conferences such as CVPR. The authors should update the references.

Additional Questions:
Is the work within the scope of the journal?: Clearly within scope

Is the manuscript technically correct?: Some minor concerns that should be easily corrected without altering the contribution or conclusions

Is the technical contribution novel?: Moderate novelty, with clear extensions of existing methods/concepts

Is the technical contribution significant?: Moderate contribution, with the possibility of an impact on the field

Is the length of the manuscript appropriate to the contribution?: Length is appropriate

Are the references appropriate, without any significant omissions?: Some significant omissions that may have a moderate impact on the novelty of the submission

Are there any references that do not appear to be relevant?: All references are directly relevant to the contribution of the manuscript

Is the manuscript properly structured and clearly written?: Some minor structural, language, or other issues of exposition that would be easily rectified

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.
: No specific references are suggested.